Separable Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Order and verifying solutions**

State the order of the differential equation $y'' + 3y' - 2y = \sin x$.

Second order (the highest derivative appearing is y'').

2 **[Foundational]** • **Target: Order and verifying solutions**

State the order of $\left(\frac{dy}{dx}\right)^3 + xy = 0$, and note that order depends only on the highest derivative, not on any power of it.

First order — the equation involves only y' (cubed), no higher derivative.

3 **[Foundational]** • **Target: Order and verifying solutions**

Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.

$y' = 2e^{2x}$, so $y' - 2y = 2e^{2x} - 2e^{2x} = 0$. Yes, it is a solution.

4 **[Foundational]** • **Target: Order and verifying solutions**

Verify that $y = \frac{1}{x}$ is a solution of $y' + y^2 = 0$ for $x \neq 0$.

$y' = -\frac{1}{x^2}$ and $y^2 = \frac{1}{x^2}$, so $y' + y^2 = -\frac{1}{x^2} + \frac{1}{x^2} = 0$. Yes, it is a solution.

5 **[Foundational]** • **Target: Recognizing and solving separable equations**

Solve $\frac{dy}{dx} = xy$.

Separate: $\frac{dy}{y} = x\,dx$. Integrate: $\ln|y| = \frac{x^2}{2} + C$. So $y = Ae^{x^2/2}$.

6 **[Foundational]** • **Target: Recognizing and solving separable equations**

Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.

Separate: $\frac{dy}{y^2} = \frac{dx}{x}$. Integrate: $-\frac{1}{y} = \ln x + C$. So $y = \frac{-1}{\ln x + C}$.

7 **[Foundational]** • **Target: Recognizing and solving separable equations**

Solve $\frac{dy}{dx} = \frac{x+1}{y}$.

Separate: $y\,dy = (x+1)\,dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + x + C$. So $y^2 = x^2 + 2x + C_1$ (renaming $2C$ as C_1).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Recognizing and solving separable equations

Solve $\frac{dy}{dx} = 6x^2y^2$.

Separate: $\frac{dy}{y^2} = 6x^2\,dx$. Integrate: $-\frac{1}{y} = 2x^3 + C$. So $y = \frac{-1}{2x^3 + C}$.

9 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium (constant) solutions of $\frac{dy}{dx} = y(3 - y)$.

Set $y(3 - y) = 0$: the equilibrium solutions are $y = 0$ and $y = 3$.

10 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

For $\frac{dy}{dx} = y(3 - y)$, determine whether $y = 0$ and $y = 3$ are stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below each.

For $0 < y < 3$: $y(3 - y) > 0$, so solutions increase away from $y = 0$ and toward $y = 3$. For $y > 3$: $y(3 - y) < 0$, so solutions decrease back toward $y = 3$. Hence $y = 0$ is unstable and $y = 3$ is stable.

11 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium solutions of $\frac{dy}{dx} = (y - 2)(y + 1)$ and classify each as stable or unstable.

Equilibria: $y = 2$ and $y = -1$. For $-1 < y < 2$, $(y - 2)(y + 1) < 0$, so solutions decrease toward $y = -1$: $y = -1$ is stable. For $y > 2$, the product is positive, pushing solutions away: $y = 2$ is unstable.

12 [Intermediate] • Target: General versus particular solutions

Find the general solution of $\frac{dy}{dx} = 2x$, then find the particular solution satisfying $y(1) = 3$.

General solution: $y = x^2 + C$ (this one is already separated, with $\frac{dy}{dx}$ depending only on x). Since $y(1) = 3$: $1 + C = 3 \Rightarrow C = 2$. Particular solution: $y = x^2 + 2$.

13 [Intermediate] • Target: General versus particular solutions

Explain, in terms of the arbitrary constant C, why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.

Integrating introduces one arbitrary constant C, so the general solution is really a one-parameter family of curves, all satisfying the ODE. An initial condition $y(x_0) = y_0$ picks out the single curve in that family passing through (x_0, y_0), which (under standard existence/uniqueness conditions) fixes C to exactly one value.

14 [Intermediate] • Target: Initial value problems

Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.

$\frac{dy}{y} = 3x^2\,dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$. Since $y(0) = 2$, $A = 2$: $y = 2e^{x^3}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Initial value problems** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.

$\frac{dy}{y} = -2x\,dx \Rightarrow \ln|y| = -x^2 + C \Rightarrow y = Ae^{-x^2}$. Since $y(0) = 5$, $A = 5$: $y = 5e^{-x^2}$.

16 **[Challenging]** • **Target: Initial value problems** • **Cross-Lesson: 5.01 Variation of Parameters**

Solve $\frac{dy}{dx} = 4x^3y$, $y(0) = -3$.

$\frac{dy}{y} = 4x^3\,dx \Rightarrow \ln|y| = x^4 + C \Rightarrow y = Ae^{x^4}$. Since $y(0) = -3$, $A = -3$: $y = -3e^{x^4}$.

17 **[Challenging]** • **Target: Initial value problems** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Solve $y' = -\frac{1}{2}y$, $y(0) = 8$.

$\frac{dy}{y} = -\frac{1}{2}\,dx \Rightarrow \ln|y| = -\frac{x}{2} + C \Rightarrow y = Ae^{-x/2}$. Since $y(0) = 8$, $A = 8$: $y = 8e^{-x/2}$.

18 **[Challenging]** • **Target: Application: radioactive decay and population growth** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k.

$A = 100e^{-kt}$. Then $50 = 100e^{-5k} \Rightarrow e^{-5k} = \frac{1}{2} \Rightarrow -5k = \ln\frac{1}{2} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$ per unit time.

19 **[Challenging]** • **Target: Application: radioactive decay and population growth** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.

$P(t) = 200e^{0.03t}$, so $P(10) = 200e^{0.3} \approx 269.97$.

20 **[Challenging]** • **Target: Application: radioactive decay and population growth** • **Cross-Lesson: 5.01 Variation of Parameters**

A bank account earns continuously-compounded interest, $\frac{dA}{dt} = 0.05A$, with $A(0) = 1000$. Find $A(20)$ and the doubling time.

$A(t) = 1000e^{0.05t}$, so $A(20) = 1000e^{1} \approx 2718.28$. Doubling time solves $e^{0.05t} = 2 \Rightarrow t = \frac{\ln 2}{0.05} \approx 13.86$ years.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Separation Form and Integrals

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is separable if it can be factored into:

a. $\frac{dy}{dx} = g(x)h(y)$ ✓

b. $\frac{dy}{dx} = g(x) + h(y)$

c. $\frac{dy}{dx} = g(x/y)$

d. $\frac{dy}{dx} = x^n + y^n$

A differential equation is separable if the right-hand side factors into a product of a function of x alone and a function of y alone: $g(x)h(y)$.

22 [Foundational] • Target: Separation Form and Integrals

Separate variables for $\frac{dy}{dx} = 3x^2y^2$.

a. $\frac{1}{y^2}dy = 3x^2dx$ ✓

b. $y^2dy = 3x^2dx$

c. $\frac{1}{3x^2}dy = y^2dx$

d. $dy = 3x^2y^2dx$

Divide both sides by y^2 and multiply by dx: $\frac{1}{y^2}dy = 3x^2dx$.

23 [Foundational] • Target: Separation Form and Integrals

Solve the general equation $\frac{1}{y^2}dy = 3x^2dx$.

a. $y = -\frac{1}{x^3 + C}$ ✓

b. $y = \frac{1}{x^3 + C}$

c. $y = -x^3 + C$

d. $y = \sqrt{x^3 + C}$

Integrate both sides: $-\frac{1}{y} = x^3 + C \Rightarrow y = -\frac{1}{x^3 + C}$.

24 [Foundational] • Target: Initial Value Problems (IVPs)

Solve the IVP: $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = -3$.

a. $y = -\sqrt{x^2 + 9}$ ✓

b. $y = \sqrt{x^2 + 9}$

c. $y = -\sqrt{x^2 - 9}$

d. $y = -x - 3$

$ydy = xdx \Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = -3$: $(-3)^2 = 0 + 2C \Rightarrow 2C = 9$. Thus $y^2 = x^2 + 9$. Since $y(0) = -3 < 0$, we select the negative branch $y = -\sqrt{x^2 + 9}$.

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25 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dx} = 2y$ with $y(0) = 7$.

a. $y = 7e^{2x}$ ✓

b. $y = 7e^{-2x}$

c. $y = 2e^{7x}$

d. $y = 7 + 2x$

$\frac{1}{y}dy = 2dx \Rightarrow \ln|y| = 2x + C \Rightarrow y = Ae^{2x}$. Using $y(0) = 7$: $y = 7e^{2x}$.

26 [Foundational] • Target: Initial Value Problems (IVPs)

What is a singular solution of a differential equation?

a. **A solution that cannot be obtained from the general family for any value of the arbitrary constant** C ✓

b. A solution that blows up to infinity everywhere

c. A solution where the derivative is undefined

d. A solution with no arbitrary constants

A singular solution (often an equilibrium solution where $h(y) = 0$) cannot be generated by tuning the parameter C in the general solution formula.

27 [Foundational] • Target: Exponential and Logistic Growth Models

What is the differential equation for logistic population growth with carrying capacity K and growth rate r?

a. $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$ ✓

b. $\frac{dP}{dt} = r(K - P)$

c. $\frac{dP}{dt} = rP^2$

d. $\frac{dP}{dt} = \frac{rP}{K}$

The logistic differential equation is $\frac{dP}{dt} = rP(1 - P/K)$, where growth slows down as population P approaches carrying capacity K.

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28 [Foundational] • Target: Exponential and Logistic Growth Models

At what population size does the logistic growth rate $\frac{dP}{dt}$ reach its maximum?

a. $P = \frac{K}{2}$ ✓

b. $P = K$

c. $P = 0$

d. $P = \frac{K}{4}$

The quadratic $f(P) = rP - \frac{r}{K}P^2$ reaches its maximum at vertex $P = -\frac{r}{2(-r/K)} = \frac{K}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Exponential and Logistic Growth Models

What is $\lim_{t\to\infty} P(t)$ in a logistic model if $P(0) > 0$?

a. K ✓

b. ∞

c. 0

d. rK

For any positive initial population, the population asymptotically approaches the stable equilibrium carrying capacity K as $t \to \infty$.

30 [Intermediate] • Target: Newton's Law of Cooling

Newton's Law of Cooling states that the rate of change of temperature T is proportional to the difference between T and the ambient temperature T_m:

a. $\frac{dT}{dt} = -k(T - T_m)$ **with** $k > 0$ ✓

b. $\frac{dT}{dt} = -kT$

c. $\frac{dT}{dt} = k(T + T_m)$

d. $\frac{dT}{dt} = -k(T - T_m)^2$

The rate of cooling is proportional to the temperature gradient with the surroundings: $\frac{dT}{dt} = -k(T - T_m)$.

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31 [Intermediate] • Target: Newton's Law of Cooling

A metal bar at 100°C is placed in a room at 20°C. If $T(t) = 20 + 80e^{-kt}$ and cools to 60°C after 10 minutes, find e^{-10k}.

a. $\frac{1}{2}$ ✓

b. $\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

$60 = 20 + 80e^{-10k} \Rightarrow 40 = 80e^{-10k} \Rightarrow e^{-10k} = \frac{40}{80} = \frac{1}{2}$.

32 [Intermediate] • Target: Newton's Law of Cooling

Using $e^{-10k} = \frac{1}{2}$ from above, what is the temperature after 20 minutes?

a. 40°C ✓

b. 30°C

c. 50°C

d. 25°C

After 20 minutes (two 10-minute half-cooling intervals): $e^{-20k} = (e^{-10k})^2 = (1/2)^2 = 1/4$. Temperature $T(20) = 20 + 80(1/4) = 20 + 20 = 40^\circ\text{C}$.

33 [Intermediate] • Target: Newton's Law of Cooling

A hot object cools from $100\,^\circ\text{C}$ to $80\,^\circ\text{C}$ in 10 minutes in an environment kept at $20\,^\circ\text{C}$. Under Newton's Law of Cooling, at what time t is the instantaneous rate of cooling $|dT/dt|$ exactly half of its initial value at $t = 0$?

a. $10\frac{\ln 2}{\ln(4/3)}$ ✓

b. $10\frac{\ln(4/3)}{\ln 2}$

c. 20 minutes

d. $5\frac{\ln 2}{\ln(4/3)}$

Separable: $\frac{dT}{dt} = -k(T - 20) \Rightarrow |T'(t)| = k(T(t) - 20) = 80ke^{-kt}$. The initial cooling rate is $80k$. The rate is halved when $e^{-kt} = 1/2 \Rightarrow t = \frac{\ln 2}{k}$. From $T(10) - 20 = 60 = 80e^{-10k}$, we find $e^{-10k} = \frac{3}{4} \Rightarrow k = \frac{1}{10}\ln(4/3)$. Thus $t = 10\frac{\ln 2}{\ln(4/3)}$.

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34 **[Intermediate]** • **Target: Order and verifying solutions**

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.

a. 1

b. **2** ✓

c. 6

d. 12

The given differential equation is $y'' - 5y' + 6y = 12x + 4$. \\ We are given that a particular solution is of the form $y_p(x) = Ax + B$. \\ To find the coefficients A and B, we substitute $y_p(x)$ and its derivatives into the differential equation. \\ The first derivative is $y_p'(x) = A$. \\ The second derivative is $y_p''(x) = 0$. \\ Substituting these into the differential equation: \\ xBy comparing the coefficients of x on both sides, we get: \\ $6A = 12 \Rightarrow A = 2$By comparing the constant te.

35 **[Intermediate]** • **Target: Recognizing and solving separable equations**

An object cools in a room kept at a constant temperature T_a. At $t = 0$, its temperature is $100\,^\circ\text{C}$; at $t = 10$ min, it is $60\,^\circ\text{C}$; and at $t = 20$ min, it is $40\,^\circ\text{C}$. Assuming Newton's Law of Cooling, what is the room temperature T_a?

a. $20\,^\circ\text{C}$ ✓

b. $15\,^\circ\text{C}$

c. $25\,^\circ\text{C}$

d. $18\,^\circ\text{C}$

Separable: $\frac{dT}{dt} = -k(T - T_a) \Rightarrow T(t) - T_a = (100 - T_a)e^{-kt}$. At $t = 10$, $60 - T_a = (100 - T_a)e^{-10k}$. At $t = 20$, $40 - T_a = (100 - T_a)e^{-20k} = (100 - T_a)(e^{-10k})^2$. Thus $\frac{40 - T_a}{60 - T_a} = e^{-10k} = \frac{60 - T_a}{100 - T_a}$. Cross-multiplying gives $(60 - T_a)^2 = (40 - T_a)(100 - T_a) \Rightarrow 3600 - 120T_a + T_a^2 = 4000 - 140T_a + T_a^2 \Rightarrow 20T_a = 400 \Rightarrow T_a = 20\,^\circ\text{C}$.

36 **[Intermediate]** • **Target: Equilibrium solutions and qualitative behavior**

Hot tea cools from $95\,^\circ\text{C}$ to $80\,^\circ\text{C}$ in 5 minutes in a room kept at $20\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool from $80\,^\circ\text{C}$ to $50\,^\circ\text{C}$?

a. $10\frac{\ln 2}{\ln(5/4)}$

b. $5\frac{\ln(4/5)}{\ln 2}$

c. $5\frac{\ln 2}{\ln(5/4)}$ ✓

d. $5\frac{\ln(5/2)}{\ln(5/4)}$

Separable: $T(t) - 20 = 75e^{-kt}$. At $t = 5$, $80 - 20 = 60 = 75e^{-5k} \Rightarrow e^{-5k} = \frac{4}{5} \Rightarrow k = \frac{1}{5}\ln(5/4)$. For $T(t) = 50$, $30 = 75e^{-kt} \Rightarrow e^{-kt} = \frac{2}{5}$, so $t = \frac{\ln(5/2)}{k} = 5\frac{\ln(5/2)}{\ln(5/4)}$. The additional time is $t - 5 = 5\left(\frac{\ln(5/2)}{\ln(5/4)} - 1\right) = 5\frac{\ln(5/2) - \ln(5/4)}{\ln(5/4)} = 5\frac{\ln 2}{\ln(5/4)}$.

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37 [Intermediate] • Target: General versus particular solutions

A metal ingot at $800\,^\circ$C is immersed in a cooling bath kept at $50\,^\circ$C. After 1 minute, its temperature drops to $500\,^\circ$C. Under Newton's Law of Cooling, how many additional minutes will it take to cool to $100\,^\circ$C?

a. $\frac{3\ln 2}{\ln(5/3)}$

b. $\frac{\ln 3}{\ln(5/3)}$

c. $\frac{2\ln 3}{\ln(5/3)}$ ✓

d. $\frac{\ln 15}{\ln(5/3)}$

Separable: $T(t) - 50 = 750e^{-kt}$. At $t = 1$, $450 = 750e^{-k} \Rightarrow e^{-k} = \frac{3}{5} \Rightarrow k = \ln(5/3)$. For $T(t) = 100$, $50 = 750e^{-kt} \Rightarrow e^{-kt} = \frac{1}{15} \Rightarrow t = \frac{\ln 15}{\ln(5/3)}$. The additional time required is $t - 1 = \frac{\ln 15 - \ln(5/3)}{\ln(5/3)} = \frac{\ln(15 \cdot 3/5)}{\ln(5/3)} = \frac{\ln 9}{\ln(5/3)} = \frac{2\ln 3}{\ln(5/3)}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Initial value problems • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

An object cools in an environment whose ambient temperature rises linearly according to $T_m(t) = 10 + 2t$ (in $^\circ$C). Newton's Law of Cooling gives $T' = -k(T - T_m(t))$ with $k = 1$ hr^{-1}. If $T(0) = 50\,^\circ$C, what is $T(1)$?

a. $10 + 42e^{-1}$ ✓

b. $10 + 40e^{-1}$

c. $12 + 42e^{-1}$

d. $8 + 42e^{-1}$

Linear ODE: $T' + T = 10 + 2t$. Integrating factor: $\mu = e^t$. Multiplying gives $(Te^t)' = (2t + 10)e^t$. Integrating by parts yields $Te^t = (2t + 8)e^t + C \Rightarrow T(t) = 2t + 8 + Ce^{-t}$. Using $T(0) = 50$ yields $50 = 8 + C \Rightarrow C = 42$. At $t = 1$, $T(1) = 2(1) + 8 + 42e^{-1} = 10 + 42e^{-1}$.

39 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A cup of coffee at $90\,^\circ$C is placed in a room held at $20\,^\circ$C. After 10 minutes, the temperature of the coffee is $60\,^\circ$C. Assuming Newton's Law of Cooling, at what time t (in minutes) will its temperature reach $30\,^\circ$C?

a. $10\frac{\ln 7}{\ln(4/7)}$

b. $20\frac{\ln 7}{\ln(7/4)}$

c. $10\frac{\ln 7}{\ln(7/4)}$ ✓

d. $10\frac{\ln 4}{\ln(7/4)}$

Separable: $\frac{dT}{dt} = -k(T - 20) \Rightarrow T(t) = 20 + 70e^{-kt}$. At $t = 10$, $60 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \frac{4}{7}$. For $T(t) = 30$, $30 = 20 + 70e^{-kt} \Rightarrow e^{-kt} = \frac{1}{7}$. Taking logs: $-kt = -\frac{t}{10}\ln(7/4) = -\ln 7 \Rightarrow t = 10\frac{\ln 7}{\ln(7/4)}$.

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40 [Challenging] • Target: Application: Newton's Law of Cooling • Cross-Lesson: 5.01 Variation of Parameters

A thermometer initially reading $25\,^\circ\text{C}$ is taken outside into air at constant temperature T_m. After 1 minute it reads $15\,^\circ\text{C}$, and after 2 minutes it reads $9\,^\circ\text{C}$. Assuming Newton's Law of Cooling, what is the outdoor temperature T_m?

a. $-5\,^\circ\text{C}$

b. $5\,^\circ\text{C}$

c. $3\,^\circ\text{C}$

d. $0\,^\circ\text{C}$ ✓

Separable: $T(t) - T_m = (25 - T_m)e^{-kt}$. At $t = 1$, $15 - T_m = (25 - T_m)e^{-k}$. At $t = 2$, $9 - T_m = (25 - T_m)e^{-2k}$. Squaring the first ratio gives $(15 - T_m)^2 = (25 - T_m)(9 - T_m) \Rightarrow 225 - 30T_m + T_m^2 = 225 - 34T_m + T_m^2 \Rightarrow 4T_m = 0 \Rightarrow T_m = 0\,^\circ\text{C}$.

41 [Challenging] • Target: Further practice: separable equations • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Under Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_m)$ with surrounding temperature $T_m = 20\,^\circ\text{C}$. An object starts at $T(0) = 80\,^\circ\text{C}$ and cools to $T(1) = 50\,^\circ\text{C}$ at $t = 1$ hour. What is the cooling constant k?

a. $-\ln 2$

b. $\ln 2$ ✓

c. $2\ln 2$

d. $\ln(5/8)$

Separable: $\frac{dT}{T-20} = -k\,dt \Rightarrow T(t) - 20 = (80 - 20)e^{-kt} = 60e^{-kt}$. At $t = 1$, $50 - 20 = 60e^{-k} \Rightarrow 30 = 60e^{-k} \Rightarrow e^{-k} = \frac{1}{2} \Rightarrow k = \ln 2$.

42 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1 - x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

a. **0** ✓

b. 1

c. $\frac{\pi}{2}$

d. Undefined

The given differential equation is $\frac{dy}{dx} = \frac{x^2\cos(y)}{1 - x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. We are asked to find $y'(0)$. The differential equation directly gives us the expression for $\frac{dy}{dx}$, which is $y'(x)$. To find $y'(0)$, we substitute $x = 0$ and $y = y(0) = \frac{\pi}{2}$ into the differential equation: $y'(0) = \frac{(0)^2\cos(y(0))}{1 - (0)^3}$ Substitute $y(0) = \frac{\pi}{2}$: $y'(0) = \frac{0 \cdot \cos(\frac{\pi}{2})}{1 - 0}$ We know that.

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43 [Challenging] • Target: Separation Form and Integrals • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

a. $\frac{1}{2}$ ✓

b. $\frac{1}{4}$

c. $\frac{1}{2}\sin(2x)$

d. $\frac{1}{4}\cos(2x)$

The given differential equation is $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. The associated homogeneous equation is $\frac{d^2y}{dx^2} + 4y = 0$. The characteristic equation is $r^2 + 4 = 0$, which has roots $r = \pm 2i$. Thus, the fundamental set of solutions for the homogeneous equation is $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. We use the method of variation of parameters to find a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$. The Wronskian of y_1 and y_2 is.

44 [Challenging] • Target: Initial Value Problems (IVPs) • Cross-Lesson: 5.01 Variation of Parameters

Consider a second-order linear non-homogeneous differential equation of the form $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$. If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions to the corresponding homogeneous equation $y''(x) + p(x)y'(x) + q(x)y(x) = 0$, and the Wronskian is $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$, which of the following expressions represents a particular solution $y_p(x)$ obtained using the method of variation of parameters?

a. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$ ✓

b. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

c. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

d. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

The method of variation of parameters for a second-order linear non-homogeneous differential equation $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$, where $y_1(x)$ and $y_2(x)$ are linearly independent solutions to the homogeneous equation, seeks a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$. The standard formulas for $u_1'(x)$ and $u_2'(x)$ are derived as follows: $u_1'(x) = -\frac{y_2(x)g(x)}{W(y_1,y_2)(x)}$ $u_2'(x) = \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}$ Integrating these expressions gives.

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45 [Challenging] • Target: Exponential and Logistic Growth Models • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.

a. $A = 2$ ✓

b. $A = 3$

c. $A = 6$

d. $A = 12$

The given nonhomogeneous differential equation is $y'' - 5y' + 6y = 12x + 8$. The form of the nonhomogeneous term is $g(x) = 12x + 8$, which is a polynomial of degree 1. Thus, we assume a particular solution of the form $y_p(x) = Ax + B$. We need to find the first and second derivatives of $y_p(x)$: $y_p'(x) = A$ $y_p''(x) = 0$ Substitute these into the differential equation: $0 - 5(A) + 6(Ax + B) = 12x + 8$ $-5A + 6Ax + 6B = 12x + 8$ $6Ax + (6B - 5A) = 12x + 8$ Equating the coefficients of like powers of x: For.

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